Arham Hassan

+1613-985-8023 · 23cr8@queensu.ca · 64 Regent St, Kingston ON
linkedin.com/in/arham-hassan-a21457242 · https://github.com/ar7mflow

AI / ML ENGINEERING INTERN

Third-year Computer Science (AI) student at Queen's with dedicated hands-on experience shipping AI governance systems and prototyping LLM-powered tools. SAP Certified Generative AI Developer with a track record of turning ideas into working demos, managing cross-functional teams, and executing under tight deadlines. Exceptionally passionate about building real, responsible AI systems that solve real-world problems, from enterprise compliance to education access in underserved communities.

EDUCATION

Queen's University **Expected June 2029**

B.Comp (Honours), Computer Science (AI)

Relevant Coursework:

Artificial Intelligence, Algorithms I, Software Architecture, Data Structures, Programming Paradigms, Logic for Computing Science, Discrete Structures II, Cognition & Computation, Software Specifications

External Coursework:

Java for Programmers (Athabasca University, Grade: A)

TECHNICAL SKILLS

Languages: Python, Java, C/C++, SQL, Rust

AI/ML: SAP AI Core/Launchpad, OpenAI/Anthropic APIs, prompt engineering, RAG, embeddings, function calling, PyTorch (learning), OpenCV (basic), LangChain/LlamaIndex

Tools & Frameworks: Git, FastAPI, Streamlit, Docker (learning), Figma, SAP GenAI Hub

Concepts: LLM governance & safety, compliance validation, agent workflows, computer vision, distributed systems

RELEVANT EXPERIENCE

AI Product Builder (Independent Projects) **Sep 2024 - Present**

Halluci-NOT — LLM Governance & Compliance Layer for SAP GenAI

LLM governance and compliance wrapper for SAP GenAI that detects and blocks PII exposure, hallucinated SAP terms, and policy violations using a two-layer validation + severity scoring with audit-ready outputs.

Accomplishments:

- **Built two-layer validation** system (regex pattern matching + GPT-4o mini reasoning) to detect PII leaks, hallucinations, and policy violations in SAP AI workflows; achieved 100% catch rate on test scenarios with zero false positives across PII exposure, fake transaction codes, and unapproved vendor mentions
- **Designed severity scoring system** (0-100) mapping business risk to automated decisions: PASS (<30), REVIEW (31-70), BLOCK (71-100); generated structured JSON audit logs for GDPR compliance and governance teams
- **Validated against real SAP transaction codes** (ME21N, EKKO, ME23N) and caught fabricated commands embedded in natural language; prevented operational failures from confident AI hallucinations that could halt procurement workflows
- **Implemented entirely in SAP AI Launchpad** using prompt engineering, function calling, and allowlist verification (15 T-codes, 7 modules, 7 tables, 4 approved vendors); documented v2 roadmap with persistent logging, feedback loops, and SAP BTP integration
- **Tech:** Python, SAP AI Core/Launchpad, OpenAI GPT-4o mini, prompt engineering, compliance validation, JSON output contracts
- **Code:** github.com/arhamhassan/hallucinot

Zahan ED — Offline AI Tutor for Better Education (Founder, Early-Stage R&D) **Apr 2025 - Present**

Founder-led R&D building an offline-first AI tutoring tool for low-resource classrooms, with pilot planning in Pakistan; scoping the MVP architecture (quantized on-device LLM inference, curriculum RAG, and quiz-based learning analytics) and validating requirements through teacher/student interviews.

Accomplishments:

- Researching offline LLM deployment for low-resource schools in Pakistan; scoping MVP requirements including quantized models (Llama 3.2, Mistral), curriculum RAG, and learning impact metrics through interviews with teachers and students in rural communities
- Designed initial product wireframes and landing page; investigating model quantization for <8GB RAM deployment and context-aware tutoring flows with spaced repetition to maximize learning retention in resource-constrained environments
- Upskilling in enterprise AI deployment patterns via SAP Business AI coursework to understand governance frameworks, evaluation methodologies, and production-grade integration for education technology at scale

Queen's VEX U Robotics Team — AI/Software R&D (Autonomy & Perception) **Sep 2025 - Present**

Hands-on robotics software work for a competition robot, collaborating across mechanical/electrical teams to ship dependable autonomy features under deadline-driven build cycles.

Accomplishments:

- Contributing to autonomy and perception systems for VEX U competition robot; implementing computer vision algorithms for object detection and path planning under hardware constraints and competition deadlines & Rust API's.
- Collaborating with mechanical and electrical teams to integrate sensor data and optimize real-time decision-making for autonomous navigation challenges
- Tech: Python, Rust, OpenCV, sensor fusion, embedded systems programming

LEADERSHIP & OTHER SKILLS (SELECTED)

Queen's Midnight Club & Other Clubs — Marketing/Brand Building Manager **Sep 2023 - Present**

- Led end-to-end content production pipeline (concept → shoot → edit → publish) for automotive club events; coordinated sponsor asset delivery and managed volunteer teams for 15+ events annually
- Built lightweight approval workflows and content calendars to ensure consistent brand voice and on-time delivery for partners and social media campaigns

ASUS (Arts & Science Undergraduate Society) — Videography Director **Sep 2024 - Apr 2025**

- Managed videography team for Queen's largest student-run society (30+ clubs, 500+ events/year); planned, filmed, and edited promotional content, event recaps, and campaign videos with consistent brand alignment and high production value
- Coordinated with event organizers, marketing teams, and executives to deliver content under tight turnaround times while maintaining creative standards across platforms

RELATED EXTRA PROJECTS

Makers Faire — Autonomous Quadcopter (Team Lead) **2019**

Accomplishments:

- Built autonomous quadcopter prototypes from concept sketches under time pressure; tuned PID controllers for stable flight and documented step-by-step build process reused by 10+ classmates
- Ran safety briefings and established flight testing protocols; maintained a near-zero accident rate during 20+ prototype flight sessions through clear communication and risk management

CERTIFICATIONS & RECOGNITION

- **SAP Certified –** SAP Generative AI Developer (Issued Jan 2026, Expires Jan 2027)
- **SAP AI Business Solution** Courses (in progress)
- **AI Awareness Seminar Host –** Certificate of Appreciation (2020)
- **RoboNXT Robotics Program** (2019)
- **Best Delegate** (Runner-Up, UNICEF) **– Model UN** (2021)
- **WHMIS 2015** and **AODA Customer Service** Awareness/Training